DISCOVER · wide-field surveys find $\mathcal{O}(1000)$ lensed AGN

Rubin LSST
PSF FWHM 0.71–0.77″

i r g · 10-yr

Euclid
PSF FWHM 0.16–0.5″

VIS Y J H · wide

Roman
PSF FWHM 0.087–0.13″

F158 F129 F106 · HLWAS

CHARACTERIZE

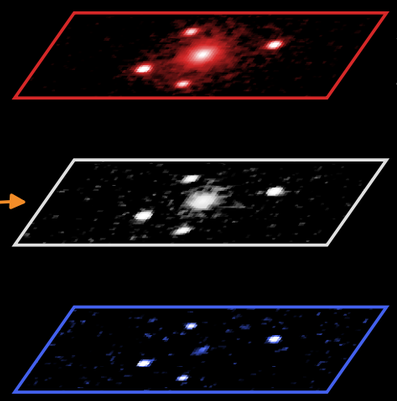
Lazuli WCC
PSF FWHM 33–62 mas

LZz LZr LZg · 4×300 s



LAZULI IFS · one pointing, every image

0.4–1.7 μm · R ≈ 100–500 · 40 mas spaxels · 3 h



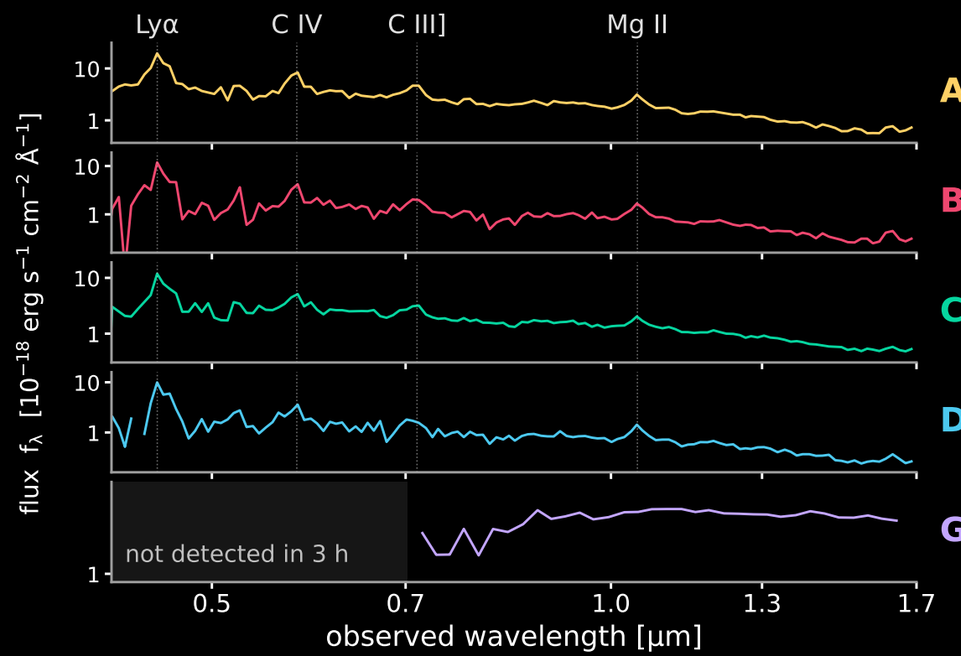
λ

1.20–1.50 μm
lens galaxy: red starlight

0.85–1.00 μm
lens galaxy appears

0.60–0.70 μm
quasar images

Same quasar lines in every image → lensing confirmed



Simulated with SLSim. Lazuli specs: Roy et al. 2026 (arXiv:2601.02556); PSF, filters, throughput, noise and exposure times are modelled, not official. Lensed-AGN counts: Oguri & Marshall 2010.